Endogenous Reachability Collapse Model (ERCM)

A Minimal Dynamical System for Loss of Recoverability Without Loss of State Space

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Abstract

We present a minimal dynamical system in which loss of recoverability appears to emerge endogenously, without removal of underlying states.

The model consists of a slow–fast dynamical system with two fast variables and one slow internal constraint variable. A thresholded memory mechanism introduces trajectory-dependent irreversibility, allowing the system to exhibit distinct dynamic regimes.

Numerical simulations suggest a sharp transition-like collapse in the fraction of initial conditions that return to a recovered regime as internal constraint increases. Importantly, this occurs without explicit removal of the recovered state from the system.

These observations are consistent with the hypothesis that loss of recoverability in complex systems may arise from internal constraint accumulation that reshapes accessibility, rather than from structural elimination of states.

The model is intentionally minimal and is presented as a falsifiable proof-of-principle.

1. Introduction

In many dynamical systems, failure is typically interpreted as the disappearance of stable states or attractors. However, it may be possible for a system to retain its underlying state space while losing the ability to access certain regions of it.

This work explores the following question:

Can a system lose dynamic access to recovery without removing the recovery state itself?

Such a mechanism may be relevant in systems where internal constraints accumulate over time, potentially altering accessibility without explicitly modifying system structure.

To investigate this, we construct a minimal dynamical model in which loss of recoverability emerges from internal dynamics rather than imposed constraints.

In contrast to trajectory-level analyses, this model focuses on ensemble-level behavior, quantifying the fraction of initial conditions that remain dynamically recoverable.

This model complements trajectory-level geometric descriptions by providing an ensemble-level measure of accessibility.

2. Model Definition

We consider a dynamical system composed of:

- Two fast variables: x(t), y(t)
- One slow internal variable: z(t), representing accumulated internal constraint (memory-like variable)

The system evolves as:

$$\dot{x} = y$$

$$\dot{y} = -\delta y + (x - x^3) - \alpha\, z \cdot \max(0,\ x) + u(t)$$

$$\dot{z} = \epsilon(\beta x^2 - z) + \gamma \cdot \max(0,\ x - x_c)$$

Where:

- z encodes internal constraint accumulation
- α controls coupling between constraint and system dynamics
- γ introduces threshold-dependent memory effects
- x_c defines a threshold for enhanced constraint accumulation
- $u(t)$ denotes an external forcing term applied consistently across simulations.
  In the simulations presented here, u(t) is implemented as a Gaussian pulse with fixed amplitude and duration.

3. Definition of Recoverability

Recoverability is operationally defined as the ability of a trajectory to return to a predefined neighborhood of a recovered state in phase space.

The recovered state corresponds to the stable fixed point of the system.

Let:

$\mathcal{R}$ = {(x,y) | (x,y) lies within a predefined neighborhood of the recovered state}

A trajectory is considered to have recovered if it enters this region within a finite time window.

The reachable fraction is defined as:

f(α) denotes the fraction of initial conditions that return to the recovered region.

4. Numerical Experiments

Simulations are performed across a grid of initial conditions in the (x, y) plane, with fixed initial constraint z(0) = 0.

For each value of the internal constraint parameter α, the fraction of trajectories returning to the recovered region is computed.

The same forcing protocol is applied across all simulations to ensure comparability of system response.

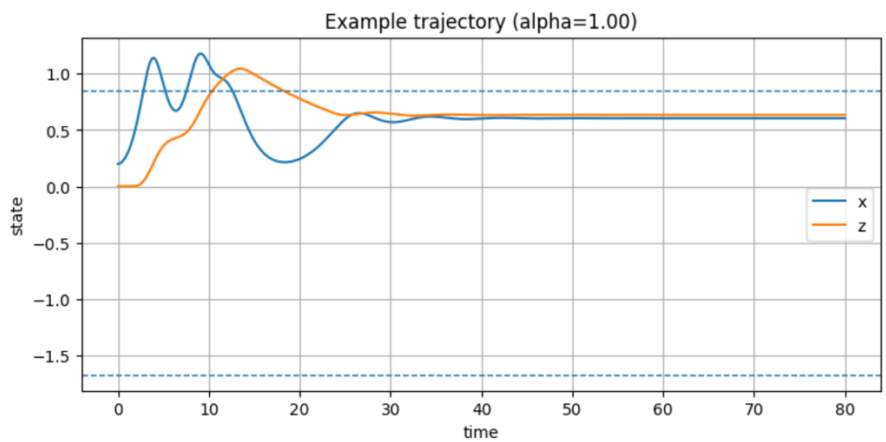

Figure 1. *Example trajectory showing system dynamics under internal constraint.*

Time evolution of the fast variable x(t) and slow internal constraint z(t) for a representative initial condition. The dashed lines indicate the recovered region.

The trajectory initially approaches the recovered region, but internal constraint accumulation alters the dynamics, illustrating how accessibility can be affected without removal of the state itself.

5. Results

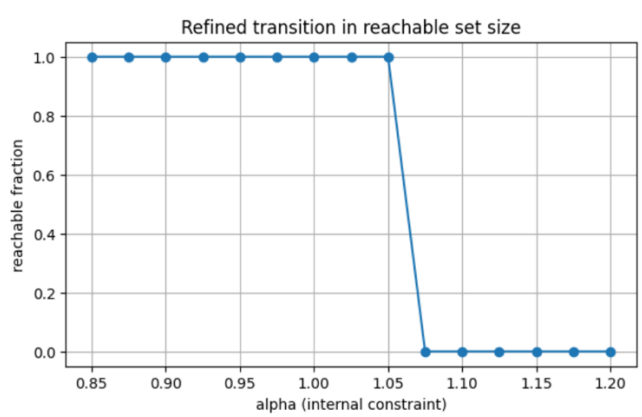

Figure 2. *Reachable fraction as a function of internal constraint strength α.*

The fraction of initial conditions that return to the recovered region is shown as α increases. The system exhibits a sharp transition from near-complete recoverability to near-zero recoverability over a narrow range of α.

This behavior is consistent with a collapse of the reachable set rather than disappearance of the recovered state.

This transition is reflected both in the fraction of returning trajectories (Figure 2) and in the contraction of the recovery basin in phase space (Figure 3).

5.1 Reachable set contraction

Results indicate a sharp transition-like collapse in the reachable fraction as α increases:

- For low values of α, the reachable fraction remains close to 1
- For higher values of α, the reachable fraction decreases rapidly

The collapse occurs within a narrow interval of the control parameter α.

5.2 Persistence of the recovered state

Numerical results are consistent with the persistence of the recovered state across the explored parameter range, although this has not been established analytically.

The observed loss of recoverability is therefore not due to disappearance of the recovered state, but rather due to loss of dynamic accessibility to it.

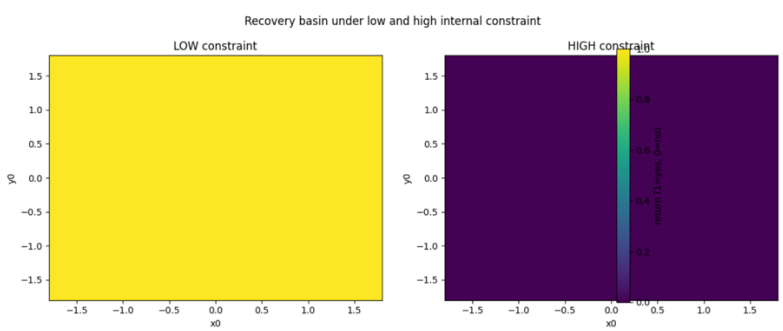

Figure 3. Recovery basin under low and high internal constraint.

Left: At low α, nearly all initial conditions return to the recovered region.
Right: At high α, the recovery basin collapses, and most trajectories fail to return.

This illustrates a contraction of the reachable set in phase space while the underlying system structure remains unchanged.

5.3 Interpretation

These observations are consistent with the following interpretation:

The system may lose access to a recovered state without that state being removed from the system.

This behavior suggests a contraction of the reachable set rather than a loss of state space.

6. Discussion

The model provides a minimal constructive example in which recoverability is reduced through internal dynamics.

The mechanism appears to involve:

- accumulation of internal constraint
- threshold-dependent memory effects
- dynamic modulation of accessibility

This suggests that failure in certain systems may arise from internal constraint accumulation that alters accessibility rather than from structural breakdown.

Potential domains where similar mechanisms may be relevant include:

- metabolic systems
- redox dynamics
- neural systems
- complex adaptive systems

Further work is required to establish empirical grounding.

7. Limitations

This model is intentionally minimal and does not represent a specific physical system.

In particular:

- the functional form of the constraint dynamics is phenomenological
- no direct empirical mapping is established
- topological properties of the flow are not formally analyzed
- the transition-like behavior is based on numerical simulations and has not been established analytically

The results should therefore be interpreted as a proof-of-principle.

8. Conclusion

We have presented a minimal dynamical system in which:

- the state space remains intact
- a recovered state persists
- but access to that state becomes progressively limited due to internal dynamics

These results are consistent with the hypothesis that:

Loss of recoverability may arise from loss of accessibility within the state space, rather than from the disappearance of states themselves.